

Microsoft Excel 16.0 Answer Report

Worksheet: [A10 SolverAssignment Spring25 NH.xlsx]Sheet1

Report Created: 2/22/2025 11:25:55 PM

Result: Solver found a solution. All Constraints and optimality conditions are satisfied.

Solver Engine

Solver Options

Objective Cell (Min)

| Cell   | Name       | Original Value | Final Value |
|--------|------------|----------------|-------------|
| \$G\$7 | Total Cost | 0              | 2           |

Variable Cells

| Cell          | Name | Original Value | Final Value | Integer |
|---------------|------|----------------|-------------|---------|
| \$H\$3:\$H\$6 |      |                |             |         |

Constraints

| Cell          | Name     | Cell Value    | Formula | Status | Slack |
|---------------|----------|---------------|---------|--------|-------|
| \$C\$7:\$F\$7 | >=       | \$C\$8:\$F\$8 |         |        |       |
| \$H\$3:\$H\$6 | =Integer |               |         |        |       |

|                         | Vitamin-A   | Vegetables | Fruit     | Iron     | Cost     | Quantity |
|-------------------------|-------------|------------|-----------|----------|----------|----------|
| Gummies                 | 250         | 3          | 2         | 2        | \$0.50   | 2        |
| Tablets                 | 200         | 1          | 2         | 0        | \$0.20   | 1        |
| Capsules                | 150         | 0          | 4         | 1        | \$0.30   | 0        |
| Syrup                   | 400         | 2          | 4         | 5        | \$0.80   | 1        |
| <b>Total</b>            | <b>1100</b> | <b>9</b>   | <b>10</b> | <b>9</b> | <b>2</b> |          |
| <b>Min. Constraints</b> | <b>600</b>  | <b>7</b>   | <b>10</b> | <b>8</b> |          |          |
|                         | (in mcg)    | (in gram)  | (in gram) | (in mg)  |          |          |



*\*How to Interpret?*

*The table shows that each of the Gummies contains 250mcg of Vitamin-A, 3 grams of Vegetables, 2 grams of Fruit, and 2 mg of Iron and costs \$0.5 each.*

### **Objective:**

Consider the problem of Multivitamin Optimization.

There are four different forms of multivitamins:

Gummies, Tablets, Capsules, and Syrup. The objective is to find a minimum-cost multivitamin combination that contains at least 600 mcg of Vitamin-A, at least 7 gram of Vegetables, at least 10 gram of Fruit, and at least 8 mg of Iron.

Using this excel sheet create a model and find the optimal quantity of desserts using Solver. You should do the following:

1. Establish Decision Variables
2. Create objective function (using formula)
3. Create constraints
4. Use Solver to come up with the solution
5. Save Solver Answer Report in another sheet
6. Submit the excel file.

Use the shaded areas (yellow and green) to develop the model.



**Total Points: 32**

**1. Decision Variables (not hardcoded; generated by Solver): 4 points**

- a. Gummies (H3) = 2
- b. Tablets (H4) = 1
- c. Capsules (H5) = 0
- d. Syrup (H6) = 1

**2. Objective Function: 3 points**

Minimize=>Cost=G\*.5+T\*.2+C\*.3+S\*.8

**3. Constraints: 5 points**

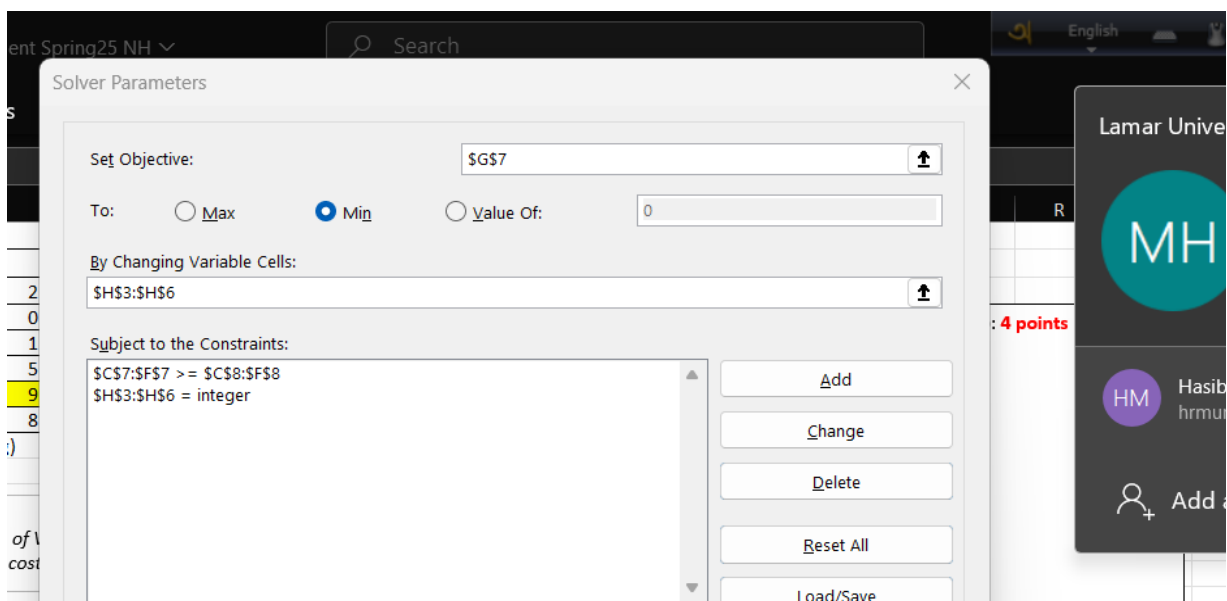
- a. Vitamin A >= 600
- b. Vegetables >=7
- c. Fruit >=10
- d. Iron >=8
- e. H3-H6 (Quantities) - Quantities are integer.

**4. Optimum quantities & Costs (not hardcoded; generated by Solver): 5 points**

- a. Vitamin A (C7) =1100
- b. Vegetables (D7) =9
- c. Fruit (E7) =10
- d. Iron (F7) =9
- e. Cost (G7) = 2

**5. Provide screenshot of Solver page (with all necessary input): 10 Points**

**6. Provide Answer Report 1 in a different sheet (generated by Solver): 5 points**





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